Attendance Code: 809762

CMP2028M

Natural Language Processing

Week 3: “Text Classification”

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Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: **Text classification**
- Week 4, 27-27 Feb: Vector semantics and embeddings
- Week 5, 5-6 Mar: Large Language Models (LLMs)
- Week 6, 12-13 Mar: Question answering & information retrieval
- Week 7, 19-20 Mar: Constituent and dependency parsing
- Week 8, 26-27 Mar: Information extraction
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: Machine translation
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

Easter break

Last Week

N-grams are probabilistic language models.

- Estimate conditional probabilities using MLE.
- Evaluate them using the metric ‘Perplexity’.
- Computing the probabilities of sentences.

$$P(\texttt{i}\,|\,\texttt{<s>}) = 0.25 \qquad P(\texttt{english}\,|\,\texttt{want}) = 0.0011$$
$$P(\texttt{food}\,|\,\texttt{english}) = 0.5 \qquad P(\texttt{</s>}\,|\,\texttt{food}) = 0.68$$

$$\begin{aligned} P(\texttt{<s> i want english food </s>}) &= P(\texttt{i}\,|\,\texttt{<s>})P(\texttt{want}\,|\,\texttt{i})P(\texttt{english}\,|\,\texttt{want}) \\ &\qquad P(\texttt{food}\,|\,\texttt{english})P(\texttt{</s>}\,|\,\texttt{food}) \\ &= 0.25 \times 0.33 \times 0.0011 \times 0.5 \times 0.68 \\ &= 0.000031 \end{aligned}$$

Today

- **Pleliminaries of Supervised Classification**

- Methods
  - N-grams
  - Logistic Regression
  - Multi-Layer Perceptrons

Supervised Classification

- It is the most standard paradigm in machine learning.

- Given dataset $D = \{\boldsymbol{x}_1, y_1, \ldots, \boldsymbol{x}_n, y_n)$ with $\boldsymbol{x}_i$ feature vectors and y_i labels (classes), find $g(x) = \arg\max_y f(\boldsymbol{x}, y)$, where $f(\boldsymbol{x}, y)$ is provided by a classification method.

- Text-based Applications:
  - Language identification
  - Sentiment analysis
  - Spam detection
  - Question answering
  - Toxicity detection, document classification, etc.

Data Splits

- Simple dual split:



- Three-fold split:



- K-fold Cross-validation: Not covered in this module.

Evaluation Metrics: Balanced Accuracy

Actual Label	Predicted Label: POSITIVE (PP)	Predicted Label: NEGATIVE (PN)
POSITIVE (P)	**True Positive (TP)**	**False Negative (FN)**
NEGATIVE (N)	**False Positive (FP)**	**True Negative (TN)**

$$BA = \frac{TPR + TNR}{2}, \text{where}$$

$$TPR = \frac{TP}{P} = \frac{TP}{TP + FN}$$

Sensitivity: recall of (P)

$$TNR = \frac{TN}{N} = \frac{TN}{TN + FP}$$

Specificity: recall of (N)

A simpler metric is called classification accuracy $CA = \frac{TP+TN}{TP+TN+FP+FN}$, but it is not as good as BA.

Evaluation Metrics: F1-Score

Predicted Label

Actual Label	POSITIVE (PP)	NEGATIVE (PN)
POSITIVE (P)	True Positive (TP)	False Negative (FN)
NEGATIVE (N)	False Positive (FP)	True Negative (TN)

$$F_1 = 2 * \left(\frac{Precision * Recall}{Precision + Recall} \right), \text{where}$$

$$Precision = \frac{TP}{PP} = \frac{TP}{TP + FP}$$ → What fraction of (PP) is actually positive?

$$Recall = \frac{TP}{P} = \frac{TP}{TP + FN}$$ → What fraction of (P) is correctly predicted as positive?

Evaluation Metrics: Example

Consider a test set with 75 positives and 925 negatives, and a classifier obtaining the following predictions:

		Predicted Label	
		POSITIVE (PP)	**NEGATIVE (PN)**
Actual Label	**POSITIVE (P)**	**TP=60**	**FN=15**
	NEGATIVE (N)	**FP=500**	**TN=425**

$$CA = \frac{TP + TN}{TP + TN + FP + FN} = \frac{485}{1000} = 0.485$$

$$BA = \frac{TPR + TNR}{2} = \frac{0.8 + 0.4594}{2} = 0.6297$$

$$Precision = \frac{TP}{PP} = \frac{60}{560} = 0.1071$$

$$Recall = \frac{TP}{P} = \frac{60}{75} = 0.8$$

$$F_1 = 2 * \left(\frac{Precision * Recall}{Precision + Recall}\right) = 0.1889$$

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- Pleliminaries of Supervised Classification

- Methods
  - **N-grams**
  - Logistic Regression
  - Multi-Layer Perceptrons

Acknowledgement: Images from https://web.stanford.edu/~jurafsky/slp3/4.pdf

N-gram Classifiers

1. Treat feature vectors and labels as word sequences:

   $x_1, x_2, \ldots, x_n, y$  $w_1, w_2, \ldots, w_{n-1}, w_n$

2. Estimate n-grams from data via bigrams $P(w_n|w_{n-1})$ or trigrams $P(w_n|w_{n-2}, w_{n-1})$ for example.

3. Compute probabilities $P(w_{1:n}) = \prod_{k=1}^{n} P(w_k|w_{k-1})$ or $P(w_{1:n}) = \prod_{k=1}^{n} P(w_k|w_{k-1}, w_{k-2})$ for the outputs (e.g., y^- and y^+)

4. Compute a probability distribution over the outputs.

N-gram Classifiers (cont.)

Compute a probability distribution over the outputs according to

$$P(y^+|w_{1:n-1}) = \frac{\exp(P(w_{1:n})^+)}{\exp(P(w_{1:n})^+) + \exp(P(w_{1:n})^-)}$$

$$P(y^-|w_{1:n-1}) = \frac{\exp(P(w_{1:n})^-)}{\exp(P(w_{1:n})^+) + \exp(P(w_{1:n})^-)}$$

The above is know as the Softmax equation … and something similar is done for more than two labels.

Today

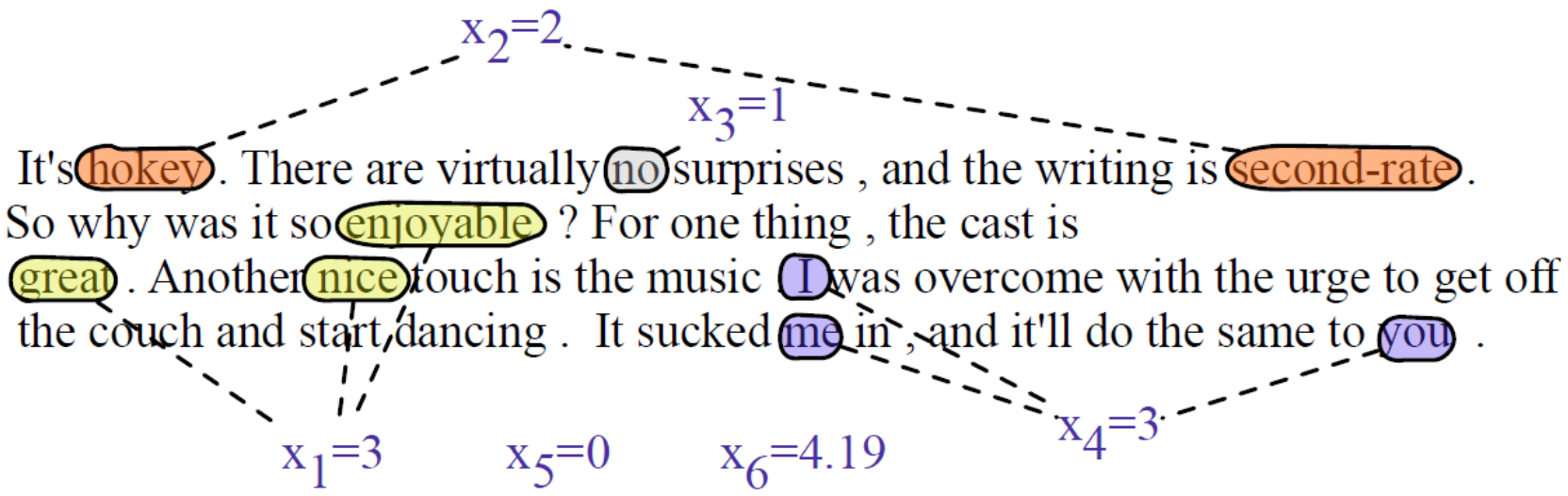
- Pleliminaries of Supervised Classification

- Methods
  - N-grams
  - **Logistic Regression**
  - Multi-Layer Perceptrons

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Example task: Sentiment Classification

Raw data:



Feature extraction:

Var	Definition	Value in Fig. 4.2
x_1	count(positive lexicon words $\in$ doc)	3
x_2	count(negative lexicon words $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if } \text{“no”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if } \text{“!”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	ln(word+punctuation count of doc)	$\ln(66) = 4.19$

Logistic Regression: In a Nutshell

Given those 6 features from the input review, the probabilities over positive & negative sentiments can be computed according to

$$
\begin{aligned}
P(+|x) = P(y=1|x) &= \sigma(\mathbf{w}\cdot\mathbf{x}+b) \\
&= \sigma([2.5,-5.0,-1.2,0.5,2.0,0.7]\cdot[3,2,1,3,0,4.19]+0.1) \\
&= \sigma(.833) \\
&= 0.70 \\
P(-|x) = P(y=0|x) &= 1-\sigma(\mathbf{w}\cdot\mathbf{x}+b) \\
&= 0.30
\end{aligned}
$$

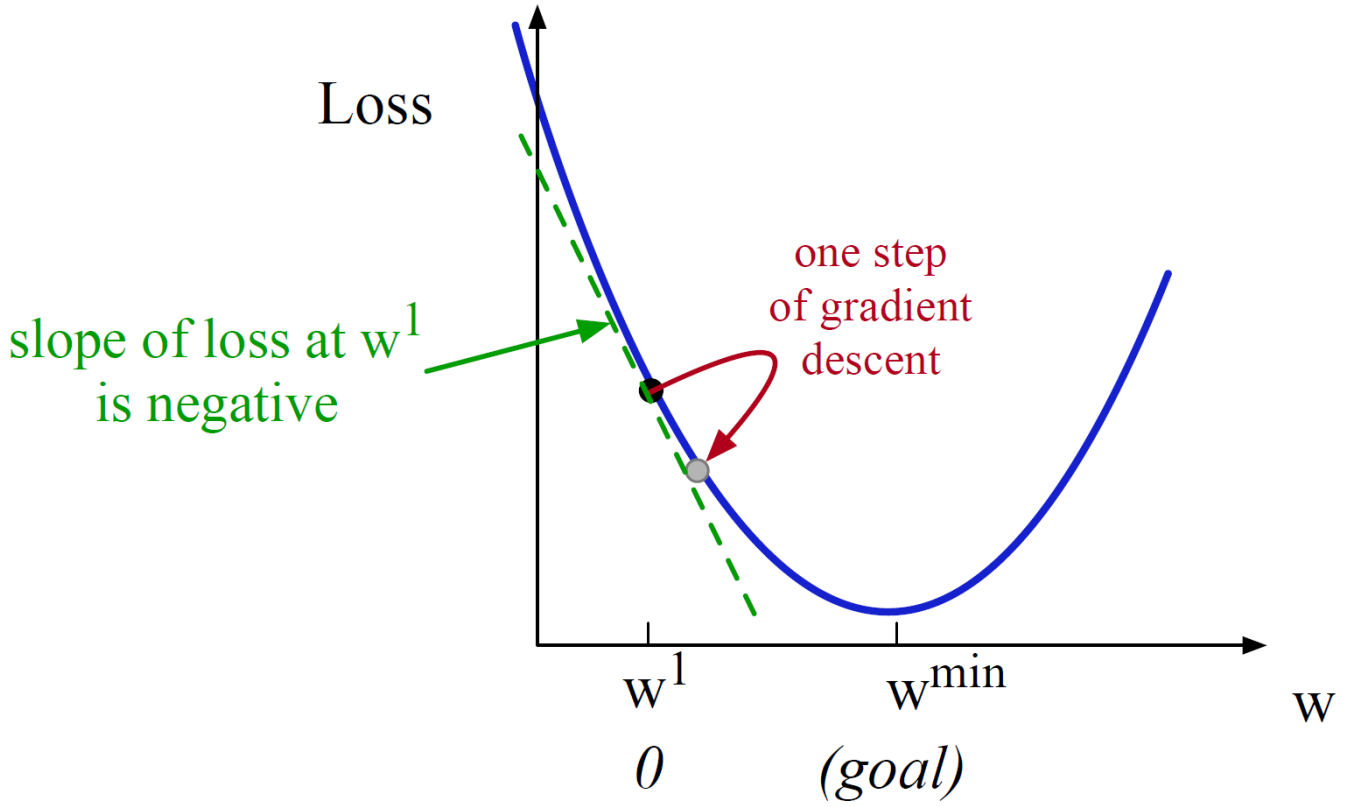
dot product

w are weights induced by a logistic regression model.

Logistic Regression: Training

Training consists in finding weights $w = w - \eta \frac{dLoss}{dw}$ that minimise the error/loss on training data:

learning rate



The procedure for finding **w** is called gradient descent.

Scaling Input Features

Data **standardisation** (in units of stdev from the mean):

$$\mu_i = \frac{1}{m}\sum_{j=1}^{m} x_i^{(j)} \qquad \sigma_i = \sqrt{\frac{1}{m}\sum_{j=1}^{m}\left(x_i^{(j)} - \mu_i\right)^2}$$

$$x_i' = \frac{x_i - \mu_i}{\sigma_i}$$

Data **normalisation** (values 0 and 1):

$$x_i' = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)}$$

They help to convert inputs to a standard range of values.

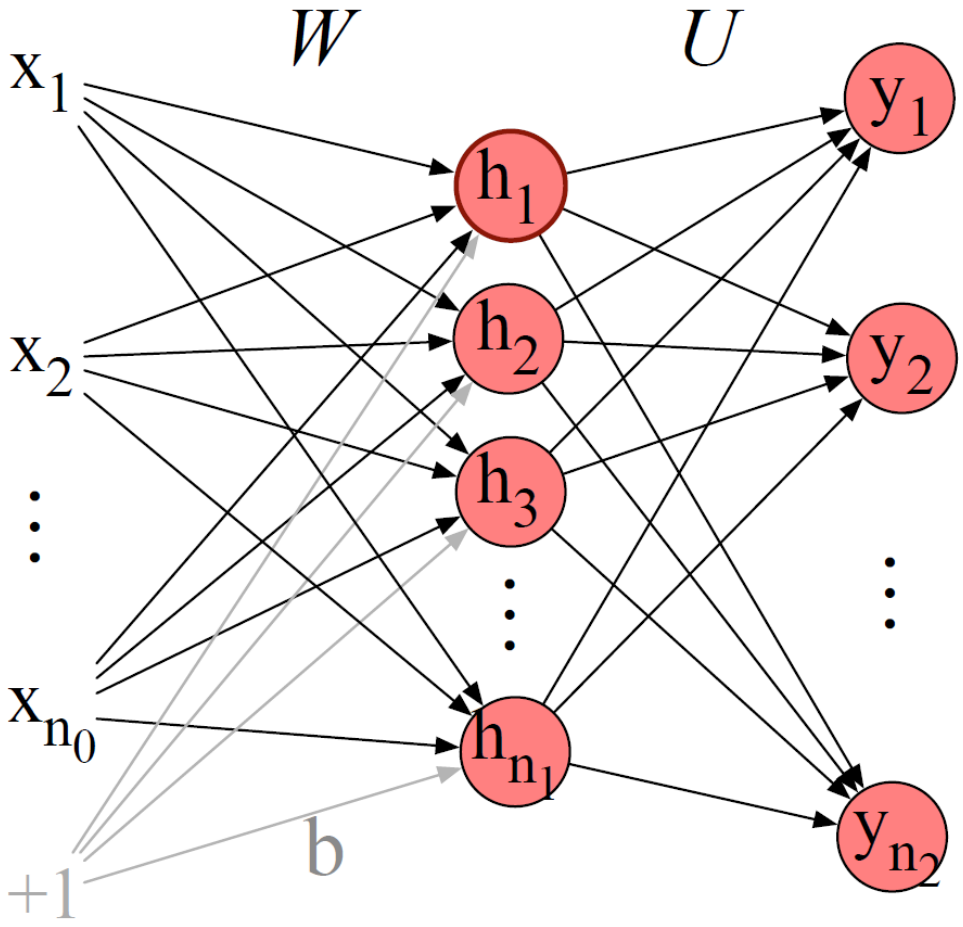
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Multi-Layer Perceptrons (MLPs)



$f()$ denotes an activation function, e.g.,

$$\sigma(net) = \frac{1}{1 + e^{-net}}$$

Each neuron calculates $f(\sum_i w_i x_i + b) = f(\boldsymbol{w} \cdot \boldsymbol{x} + b)$.

J. Lederer. "Activation Functions in Artificial Neural Networks:A Systematic Overview", CoRR, 2021.

Multi-Layer Perceptrons (cont.)

- MLPs are trained using the backpropagation algorithm—not covered in this module.

- The outputs can use the Softmax function to output a probability distribution: $$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^{K} \exp(z_j)}$$

- Whilst n-gram classifiers are fully-probabilistic models, logistic regression and MLPs are gradient-based models.

- MLPs are scalable to a large number of inputs & outputs.

Consider scaling your data when using MLPs.

Today

- **Supervised classification** is useful for a wide number of text-based applications with features $\mathbf{x} = \{x_1, x_2, \ldots, x_n\}$ and labels y. The later being binary $(+, -)$ or multiclass $(-$, neutral, $+$, etc.)

- There are many **methods** available, probabilistic and neural-based are only some of them.
  - N-gram classifiers
  - Logistic regression
  - Multi-layer perceptrons

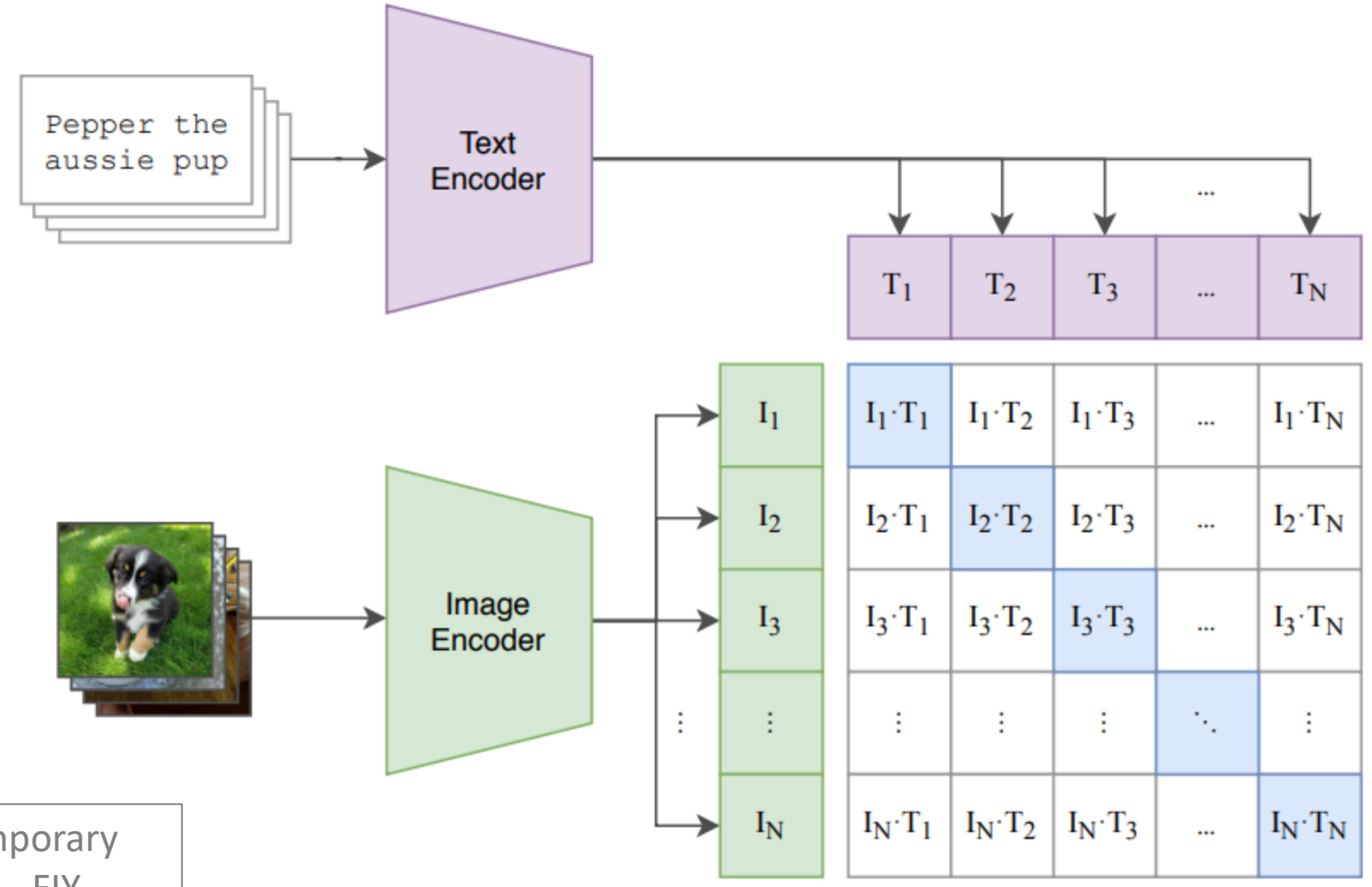
What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 4, 2026. [recommended]
  - Dreseitl, et al. "Logistic regression and ANN classification models: a methodology review", 2003

- Workshop **tomorrow**: Python code for training and evaluating text-based supervised classifiers.

- Next lecture: Vector Semantics and Embeddings.

Appendix: Multimodal Text Classification



More contemporary classification—FIY. Image source